CG2023 Signals & Systems
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Signals Classification

A signal is $x(t)$ is a **continuous-time (CTS)** is t is a continuous variable.
↳ **Analog Signal:** A CTS that can take on any value in the range (a, b) ∈ ℝ, where a and/or b may be ∞.
↳ **Digital Signal:** A CTS that can only take on a finite number of distinct displacement levels.
A signal $x(t)$ is **discrete-time (DTS)** if it is defined only at discrete time points. It takes the form of a sequence of numbers.
↳ **Analog Sequence:** A DTS that can take on any value in the range (a, b) ∈ ℝ, where a and/or b may be ∞.
↳ **Digital Sequence:** A DTS that can take on only a finite number of distinct displacement levels. Usually obtained by quantization.
Bounded Signal: A CTS $x(t)$ is bounded if there exists a $0 < M < \infty$ such that $|x(t)| \leq M;\ \forall t$.
Absolutely Integrable Signal: A CTS $x(t)$ is **absolutely integrable** if its integral from –∞ to ∞ is < ∞.
Periodic Signal: A signal $x(t)$ is periodic if there is a non-zero positive value, T, satisfying $x(t) = x(t+T);\ \forall t$. The smallest value of T is called the **fundamental period** or simply period of $x(t)$.
Aperiodic Signals: Any signals that is not periodic.
Real Signal: A signal whose value at any given instant is a real number.
Complex Signal: A signal expressed in the form of $z(t) = a(t) + jb(t)$ where $a(t)$ and $b(t)$ are either real functions or real numbers.
N Distinct Roots of a Complex Number: Express the number in polar form. Use this formula $z^{1/N} = |z|^{1/N} e^{j\left(\frac{\angle z}{N} + \frac{2k\pi}{N}\right)};\quad k = 0, 1, \ldots, N-1.$
Total Energy of a Signal: $E = \int_{-\infty}^{\infty} |x(t)|^2\, dt$
↳ A signal is said to be an **energy signal** if and only if $0 < E < \infty$.
Average Power of a Signal: $P = \lim_{\tau \to \infty} \frac{1}{2\tau} \int_{-\tau}^{\tau} |x(t)|^2\, dt$
↳ A signal is said to be a **power signal** if and only if $0 < P < \infty$.
Combining both, we get $P = \frac{E}{2 \cdot \infty}$, implying that energy signals have zero average power and power signals have infinite total energy.
!! Signals that satisfy **neither** are neither energy nor power signals !!
!! **ALL** bounded periodic signals are power signals !!

Time–Domain Operations & Dirac Impulse

Time–Scaling: Replacement of t with αt, where α is a real–value factor $0 < \alpha < 1$ = Expansion, $\alpha > 1$ = Contraction, $\alpha = -1$ = Time Reversal.
Time–Shifting: Replacement of t with $(t - \beta)$. β is a real–value factor. $\beta > 0$ = Delaying by β unit of time. $\beta < 0$ = Advancing by β unit of time.
Sum of 2 Signals: Sum the values of the two signals at each time point.
Product of 2 Signals: Multiply the values of the two signals instead.
Convolution is a mathematical way of combining two signals to form a third signal. $\int_{-\infty}^{\infty} x(\alpha) y(t-\alpha)\, d\alpha = x(t) * y(t)$
↳ $\int_{-\infty}^{\infty} x(\alpha)y(t-\alpha)\, d\alpha = \int_{-\infty}^{\infty} y(\alpha)x(t-\alpha)\, d\alpha$ (Commutative)
↳ Also Associative (A∗B)∗C and Distributive A∗(B+C)
Dirac-δ Function (Unit Impulse): View it as a limiting case of a rectangle pulse as its width approaches zero, while keeping the area constant at one. Strength of impulse = Area under impulse.
Properties of Dirac-δ Function
- Symmetry: $\delta(t) = \delta(-t)$
- Sampling: $f(t)\delta(t-t_0) = f(t_0)\delta(t-t_0)$ (Result in an impulse)
- Sifting: $\int_{-\infty}^{\infty} f(t)\delta(t-t_0)\, dt = f(t_0)$
- Replication: $x(t) * \delta(t-t_0) = x(t-t_0)$ (Result in a function)

Dirac-δ Comb Function (Sampling Function): Sum of ∞ Dirac-δ.
$$x_p(t) = x(t) * \sum_n \delta(t - Tn) = \sum_n x(t - Tn)$$
$$x_s(t) = x(t) \times \sum_n \delta(t - nT) = \sum_n x(nT)\delta(t - nT)$$

Discrete–Frequency Spectrum (Fourier Series)

A signal can be represented by a composition of sinusoidal components.
Complex Exponential: $x(t) = \mu \cdot e^{j(2\pi f_0 t + \phi)} = \mu \cdot e^{j\phi} \cdot e^{j2\pi f_0 t}$
Cosine: $x_c(t) = \mu\cos(2\pi f_0 t + \phi) = \frac{1}{2}\mu e^{j(2\pi f_0 t + \phi)} + \frac{1}{2}\mu e^{-j(2\pi f_0 t + \phi)}$
Sine: $x_s(t) = \mu\sin(2\pi f_0 t + \phi) = \frac{1}{j2}\mu e^{j(2\pi f_0 t + \phi)} - \frac{1}{j2}\mu e^{-j(2\pi f_0 t + \phi)}$
Temporal Plot is used in the time-domain, while **Spectral Plots** (Magnitude and Phase Spectrum) are used in the frequency-domain.
Fourier Series: A technique used to determine the spectra of **non-sinusoidal periodic** signals such as square wave, sawtooth wave, etc.
Complex Exponential Fourier Series: Any bounded periodic signal, $x_p(t)$ with period T_p can be represented by a sum of **harmonically** related complex sinusoids → $x_p(t) = \sum_{k=-\infty}^{\infty} c_k \cdot e^{j2\pi k f_p t}$
↳ c_k are called the **Fourier Series Coefficients** of $x_p(t)$, and they constitute the **discrete–frequency spectrum** of $x_p(t)$.
↳ $\therefore c_k = \frac{1}{T_p}\int_{t_o}^{t_o+T_p} x_p(t) e^{-j2\pi kt/T_p}\, dt, \quad k = 0, \pm 1, \pm 2, \ldots$
Trigonometric Fourier Series: Fourier Series expressed in terms of cosine and sine functions by using Euler's Formula.
↳ $x_p(t) = a_0 + 2\sum_{k=1}^{\infty}[a_k\cos(2\pi kt/T_p) + b_k\sin(2\pi kt/T_p)]$
↳ $a_k = \frac{1}{T_p}\int_{t_o}^{t_o+T_p} x_p(t)\cos(2\pi kt/T_p)\, dt = \frac{c_{-k} + c_k}{2}, \quad k \geq 0$
↳ $b_k = \frac{1}{T_p}\int_{t_o}^{t_o+T_p} x_p(t)\sin(2\pi kt/T_p)\, dt = \frac{c_{-k} - c_k}{j2}, \quad k > 0$
C_k of Dirac-δ Comb $= \frac{1}{T}\int_{-T/2}^{T/2}\left(\sum_{n=-\infty}^{\infty}\delta(x - nT)\right)e^{-i\frac{2\pi k}{T}x}dx = \frac{1}{T}$

Continuous–Frequency Spectrum (Fourier Transform)

Fourier Transform: A technique used to derive the spectrum of an **aperiodic signal** in the continuous–frequency domain.
Forward Fourier Transform: $X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft}\, dt$
Inverse Fourier Transform: $x(t) = \int_{-\infty}^{\infty} X(f)e^{j2\pi ft}\, df$
Dirichlet Conditions (Existence of Fourier Transform)
- $x(t)$ has finite number of maxima & minima in **any** finite time interval.
- $x(t)$ has finite number of discontinuities in **any** finite time interval.
- $x(t)$ is absolutely integrable. (Bounded/Finite)

↳ Top 2 are satisfied by all real-world signals of interest. Last point is satisfied by **most** energy signals but violated by **all** power signals.
- $X(f)$ constitute the **continuous–frequency spectrum** of $x(t)$.

↳ For a complex function $X(f)$, consider $|X(f)| = [X(f)X^*(f)]^{1/2}$
↳ For phase, e.g. $\angle X(f) = \angle\left[\frac{A}{\alpha + j2\pi f}\right] = \angle A - \angle(\alpha + j2\pi f)$. If it is a multiplication, you add up the phases instead.
Spectral Properties of a REAL Signal
- If $x^*(t) = x(t)$ ($x(t)$ *is real*), then $X^*(f) = X(-f)$ ($X(f)$ is conjugate symmetric).

↳ $|X(f)| = |X(-f)|$ (Even) and $\angle X(f) = -\angle X(-f)$ (*Odd*)
- If $x(t)$ is REAL and EVEN, then $X(f)$ is REAL and EVEN.
- If $x(t)$ is REAL and ODD, then $X(f)$ is IMAGINARY and ODD.

↳ The above results are **applicable to c_k** of periodic signals.
↳ The positive frequency portion of a spectrum would suffice to specify a real signal completely in the frequency domain due to the conjugate symmetry property of the spectrum.
Fourier Transform of Arbitrary Periodic Signals
The Fourier Transform can be obtained by first computing the c_k of $x(t)$ then substituting into $\mathcal{F}\{\sum_{k=-\infty}^{\infty} c_k e^{j2\pi kf_p t}\} = \sum_{k=-\infty}^{\infty} c_k\delta(f - kf_p)$.
↳ The series are represented as impulses with strength c_k.

Spectral Density and Bandwidth

Energy Spectral Density describes how the total energy of a signal is distributed in the frequency domain. $E_x(f) = |X(f)|^2$ (Joules/Hz)
Rayleigh Energy Theorem: $E = \int_{-\infty}^{\infty}|x(t)|^2\, dt = \int_{-\infty}^{\infty}|X(f)|^2\, df$
Properties of $E_x(f)$: It is a real function of f, greater than or equal to zero for all f and is an even function of f if $x(t)$ is real.
↳ |*magnitude*| is always real and non-negative.
Power Spectral Density describes how the average power of a signal is distributed in the frequency domain. As power signals have infinite total energy, it is advantageous to analyse the frequency content of a power signal using a **windowed version** of it. (Multiply with rect function)
Parseval Power Theorem:
$$P = \lim_{W \to \infty} \frac{1}{2W}\int_{-W}^{W}|x(t)|^2 dt = \int_{-\infty}^{\infty} \lim_{W \to \infty} \frac{1}{2W}|X_W(f)|^2 df$$
↳ 2W is the whole length of the windowed version of $x(t)$.
Power Spectral Density: $P_x(f) = \lim_{W \to \infty} \frac{1}{2W}|X_W(f)|^2$ (Watts/Hz)
Properties of $P_x(f)$: It is a real function of f, greater than or equal to zero for all f and is an even function of f if $x(t)$ is real.
PSD of Periodic Signals: $P_x(f) = \sum_{k=-\infty}^{\infty}|c_k|^2\delta(f - kf_p)$
Average Power of Periodic Signals: $P = \int_{-\infty}^{\infty} P_x(f)\, df = \sum_{k=-\infty}^{\infty}|c_k|^2$
Bandwidth: The bandwidth of a signal $x(t)$ is a measure of the signal's band occupancy in the frequency domain.
Bandlimited Signals comprises of both lowpass and bandpass signals.
Lowpass Signals: A signal is a **bandlimited lowpass signal** if its magnitude spectrum is **concentrated around 0 Hz**, and at the same time satisfies $|X(f)| = 0$ when $|f| > B$. When $x(t)$ is real, $|X(f)|$ is symmetric about $f = 0$. Bandwidth = Half of the signal.
Bandpass Signals: A signal is a **bandlimited bandpass signal** if its magnitude spectrum is concentrated around a non-zero center frequency f_c and at the same time satisfies $|X(f)| = 0$ when $||f| - f_c| > \frac{B}{2}$
↳ When $x(t)$ is real, $|X(f)|$ is symmetric about $f = 0$. Spectral image at f_c is assumed to be the same as the spectral image at $-f_c$ and both are symmetric about f_c and $-f_c$ respectively. Bandwidth = Full signal at f_c.
NOTE: If $f_c < B$, the signal is useless as the 2 images will overlap.
Signals with Unrestricted Band: In general, practical signals are seldom strictly bandlimited but have infinite frequency extent.
Scenario: Imagine a signal with unrestricted band propagating through a system of bandwidth B. If B is finite, the signal spectrum will be truncated and may lead to unacceptable level of signal distortion. To avoid this, B should be greater or equal to the signal bandwidth. If B approaches infinity, the system white noise will also approach infinity, thus completely masking out the signal. Therefore, it is useful to define a bandwidth measure to include only the important part of the signal.
3dB Bandwidth (Lowpass): It is defined as the frequency where $|X(f)| = |X(0)|/\sqrt{2}$ first occurs when f is **increased from 0 Hz.**
3dB Bandwidth (Bandpass): It is defined as the frequency width where $|X(f)| = |X(f_c)|/\sqrt{2}$ when f is **increased & decreased from f_c.**
By definition, $\frac{E_x(f_B)}{E_x(A)} = \frac{|X(f_B)|^2}{|X(A)|^2} = \frac{1}{2} \rightarrow$ A = 0 (Lowpass) or f_c (Bandpass)
↳ One positive f_B value for Lowpass and two (about f_c) for Bandpass.
↳ If the plot is $E_x(f)$ instead of $|X(f)|$, divide by 2 instead of $\sqrt{2}$.
1st–Null Bandwidth (Lowpass): It is defined as the frequency at which $|X(f)| = 0$ first occurs when f is **increased from 0 Hz.**
1st–Null Bandwidth (Bandpass): It is defined as the frequency width at which $|X(f)| = 0$ when f is **increased & decreased from f_c.**
NOTE: There are times when both the 3dB and 1st–Null Bandwidth are not suitable as they disposed too much of the signal.
M% Energy Containment Bandwidth: It is the smallest bandwidth that contains as least M% of the total signal energy. $E_x(f)$ plots are used.
Lowpass M%: $\int_{-B}^{B} E_x(f)\, df = \frac{M}{100} \times E$
Bandpass M%: $\int_{f_c-0.5B}^{f_c+0.5B} E_x(f)\, df = \left(\frac{M}{100} \times \frac{E}{2}\right)$ (Positive Freq Only)
M% Power Containment Bandwidth: It is the smallest bandwidth that contains as least M% of the average signal power.
Periodic (Impulses) M%: B = 0 to Af_p, where A is the smallest positive integer that satisfies $\sum_{k=-A}^{A}|c_k|^2 \geq \frac{M}{100} \times P$.

Systems Classification & s–Domain Analysis

In system studies, a system is a mathematical model of a physical process that relates the input **(excitation)** signal to the output **(response)** signal.
System with Memory and without Memory (Memoryless): A system is memoryless **(static)** if its output at a given time is dependent on only the input at that time. Otherwise, the system has memory **(dynamic)**.
Causal System: A system is causal **(non-anticipative)** if its output at the present time depends on only the **present and/or past** values of its input. It is thus **not possible** for a causal system to produce an output before an input is applied to it.
Noncausal System: A system is noncausal **(anticipative)** if its output at the present time depends on **future values** of its input.
Stable System: A system is bounded-input/bounded-output **(BIBO)** stable if for every bounded input where $|x(t)| \leq K$; $\forall t$, the system produces a bounded output where $|y(t)| \leq L$; $\forall t$, in which K and L are finite positive constants. The inputs and outputs must be bounded.
Unstable System: An unstable system is one in which not all bounded inputs lead to a bounded output.
Linear System: A system that satisfies both Additivity & Homogeneity/Scaling property (Superposition). Zero input = Zero output.
Nonlinear System: System that does violates the superposition property.
Time–Invariant System: A system is time-invariant if a time shift (delay or advance) in the input signal causes the same time shift in the output signal. $T[x(t-\tau)] = y(t-\tau)$, for any real value of τ.
Time–Varying System: One that does not satisfy the above equation.
Laplace Transform Analysis of Systems: A useful analysis tool for LTI systems is the Laplace transform. The Laplace transform of a time-domain function is defined by $\tilde{F}(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)\, e^{-st}\, dt$, where s is a **complex variable**. F(s) against s is a 3D plot.

Linear Time–Invariant Systems (LTI Systems)

Impulse Response: It is the response (output) of the system when the input is a **unit impulse**. When the input is an **arbitrary** signal, the output is the **convolution** of the input and its impulse response.
Step Response: When the input of the system is a unit step function. $o(t) = \int_{-\infty}^{t} h(\tau)\, d\tau = \mathcal{L}^{-1}\left\{\frac{1}{s}\tilde{H}(s)\right\}$ while $h(t) = \frac{d}{dt}o(t)$.
Frequency Response: $H(f) = \mathcal{F}\{h(t)\} = \int_{-\infty}^{\infty} h(t)\, e^{-j2\pi ft}\, dt$.
$Y(f) = X(f) \cdot H(f)$, since $H(f)$ is a complex function of f, $|H(f)|$ is called the magnitude response and $\angle H(f)$ is called the phase response.
Transfer Function: $\tilde{H}(s) = \mathcal{L}\{h(t)\} = \int_0^{\infty} h(t)\, e^{-st}\, dt$. $\tilde{Y}(s) = \tilde{X}(s) \cdot \tilde{H}(s)$. ∗ in time domain = × in s-domain. Usually, $H(f) = \tilde{H}(j\omega)|_{\omega = 2\pi f}$, but this condition is not always true, especially when $h(t)$ violates Dirichlet's 3rd condition of being absolutely integrable. Similar to frequency response, $\tilde{H}(j\omega)$ has magnitude and phase response.
Given a transfer function $\tilde{H}(s)$, the roots of the denominator are called poles while the roots of the numerator are called zeros. **Zeros** (O) tends to zero and **Poles** (X) tends to infinity in Pole–Zero map (s–Plane). The difference in # of Poles – # of Zeros is called the **Pole-Zero excess**.
Sinusoidal Response at Steady–State
$$x(t) = Ae^{j(2\pi f_0 t + \psi)} \longrightarrow A|H(f_0)|e^{j(2\pi f_0 t + \psi + \angle H(f_0))}$$
$$A\cos(2\pi f_0 t + \psi) \longrightarrow A|H(f_0)|\cos(2\pi f_0 t + \psi + \angle H(f_0))$$
$$A\sin(2\pi f_0 t + \psi) \longrightarrow A|H(f_0)|\sin(2\pi f_0 t + \psi + \angle H(f_0))$$
If given output steady-state sinusoidal response and asked to find input, divide output magnitude with magnitude response instead. Instead of adding the phase response, subtract from the output phase instead.

System Stability: It is defined based on the locations of the system poles and the behaviour of $h(t)$ as t tends to infinity. System zeros have NO effect on system stability but on the transient response as $t \to \infty$.

BIBO Stable: All system poles lie on the LHS of s–plane. $h(t)$ converge to zero as t tends to infinity.

Marginally Stable: At least one system pole lies on the imaginary axis and no system pole on the RHS of the s–plane. Poles lying on the imaginary axis MUST be distinct (non-repeated). $h(t)$ tends to a constant or oscillate with constant amplitude.

Unstable One: At least one system pole lies on the RHS of s–plane. $h(t)$ will "blow up" and become unbounded as t tends to infinity.

Unstable Two: At least one repeated system pole lying on the imaginary axis. $h(t)$ will "blow up" and become unbounded as t tends to infinity.

First Order System: $T\frac{dy(t)}{dt} + y(t) = Kx(t)$, $\widetilde{H}(s) = \frac{\tilde{Y}(s)}{\tilde{X}(s)} = \frac{K}{Ts+1}$

Smaller T (Time Constant) value = Faster/Steeper response.

Second Order System: $\frac{d^2y(t)}{dt^2} + 2\zeta\omega_n\frac{dy(t)}{dt} + \omega_n^2 y(t) = K\omega_n^2 x(t)$

$\widetilde{H}(s) = \frac{\tilde{Y}(s)}{\tilde{X}(s)} = \frac{K\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$. Zeta > 1 (**Overdamped**) → Poles are real and distinct. Zeta = 1 (**Critically Damped**) → Poles are real and repeated. 0 < Zeta < 1 (**Underdamped**) → Poles are a complex conjugate pair. Zeta = 0 (**Undamped**) → Poles are a pure-imaginary conjugate pair.

↳ Undamped is **marginally stable** and Underdamped is **BIBO stable**.

↳ For underdamped systems, $\widetilde{H}(s) = \frac{K\omega_n^2}{(s+\sigma)^2+\omega_d^2}$, $\sigma = \omega_n\zeta$.

$\omega_d = \omega_n\sqrt{1-\zeta^2}$. For **resonance to occur**, $\zeta < \frac{1}{\sqrt{2}}$.

↳ Due to complex poles, only underdamped and undamped systems' step and impulse response are **oscillatory**.

Bode Diagrams/Plots

A Bode plot is an **approximate visualization** of the frequency response, $\widetilde{H}(j\omega)$, of a system. It consists of straight-line segments that are constructed based on the asymptotic properties of $\widetilde{H}(j\omega)$.

A Bode plot is a combination of two plots:

- The Magnitude Plot is a plot of $|\widetilde{H}(j\omega)|_{dB} = 20\log_{10}|\widetilde{H}(j\omega)|$ dB
- The Phase Plot is a plot of $\angle\widetilde{H}(j\omega)$ in degrees.

Bode plots visualize only the positive frequency side of the frequency response. This suffices for real systems as $|\widetilde{H}(j\omega)|$ is **even** and $\angle\widetilde{H}(j\omega)$ is **odd** functions of ω, respectively.

Bode Plot for dc Gain: $H(s) = K_{dc}$

- Magnitude Plot: $|\widetilde{H}(j\omega)|_{dB} = 20\log_{10}(K_{dc})$ dB
- Phase Plot: $\angle\widetilde{H}(j\omega) = 0°$

Bode Plot for Differentiator: $H(s) = K_d s$

- Differentiator is also a system zero! Outputs the derivative of input.
- Magnitude Plot: +20 dB/decade straight line passing through $(1/K_d, 0)$ and $(1, 20\log_{10}K_d)$.
- Phase Plot: $\angle\widetilde{H}(j\omega) = 90°$

Bode Plot for Integrator: $H(s) = \frac{K_i}{s}$

- Integrator is also a system pole! Outputs the integral of input.
- Magnitude Plot: –20 dB/decade straight line passing through $(K_i, 0)$ and $(1, 20\log_{10}K_i)$.
- Phase Plot: $\angle\widetilde{H}(j\omega) = -90°$

Bode Plot for Zero Factor: $\widetilde{H}(s) = \frac{s}{z_m} + 1$; $z_m > 0$

- Magnitude Plot: $\underbrace{20\log_{10}(\omega/z_m)}_{+20\text{ dB/decade slope}}$ when $\omega \gg z_m$, 0 dB when $\omega \ll z_m$.
- Phase Plot: 0° $when\ \omega \ll z_m/10$ and 90° $when\ \omega \gg 10z_m$.

Bode Plot for Pole Factor: $\widetilde{H}(s) = \frac{1}{s/p_n+1}$; $p_n > 0$

- Magnitude Plot: $\underbrace{-20\log_{10}(\omega/p_n)}_{-20\text{ dB/decade slope}}$ when $\omega \gg p_n$, 0 dB when $\omega \ll p_n$
- Phase Plot: 0° $when\ \omega \ll p_n/10$ and $-90°$ $when\ \omega \gg 10p_n$.

Bode Plot for 2nd Order Factor: $\widetilde{H}(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$; $\omega_n > 0$, $0 \le \zeta \le 1$

- Magnitude Plot: $\underbrace{-40\log_{10}(\omega/z_m)}_{-40\text{ dB/decade slope}}$ when $\omega \gg \omega_n$, 0 dB when $\omega \ll \omega_n$
- Phase Plot: 0° $when\ \omega \ll \omega_n/10$ and $-180°$ $when\ \omega \gg 10\omega_n$.

↳ The frequency responses of the undamped, underdamped and critically damped 2nd-order systems are identical and have the same Bode straight-line plots.

Asymptotic Behaviour of Bode Plots: Assumptions → System is BIBO stable or marginally stable, and has no zero lying on the right-half s-plane

Asymptotic PHASE of Phase Plot $[\angle\widetilde{H}(j\omega)]$:

- High Freq: $\lim_{\omega\to\infty}\angle\widetilde{H}(j\omega) = [\text{No. of Poles} - No.\text{ of Zeros}] \times (-90°)$
- Low Freq: $\lim_{\omega\to 0}\angle\widetilde{H}(j\omega) = \left[\text{No. of } \int dt - \text{No. of } \frac{d}{dt}\right] \times (-90°)$

Asymptotic SLOPE of Magnitude Plot $\left[|\widetilde{H}(j\omega)|_{dB}\right]$:

- High Freq: [No. of Poles – No. of Zeros] × (–20 dB/decade)
- Low Freq: $\left[\text{No. of } \int dt - \text{No. of } \frac{d}{dt}\right] \times (-20\text{ dB/decade})$

↳ Limit s → ∞ or 0 in the transfer function to know $\widetilde{H}(s)$ behaviour.

Resonance in Second Order Systems

- Magnitude response has a 'hump' around $\omega = \omega_n$. Magnitude of the 'hump' is called resonant peak and the frequency at which it occurs is called resonant frequency.
- As ζ tends to zero, the resonant peak tends to infinity and the resonant frequency tends to ω_n.
- Resonance does not exist when ζ is increased beyond a certain value.
- **Resonant Frequency:** $\omega_r = \omega_n\sqrt{1-2\zeta^2}$
- **Resonant Peak:** $|\widetilde{H}(j\omega_r)| = \frac{K}{2\zeta\sqrt{1-\zeta^2}}$, NOT in dB!

↳ Valid only for $\zeta^2 < 0.5$ since ω_r must be real. No resonance when $\zeta^2 \ge 0.5$.

Deriving System Transfer function from Bode Magnitude Plot

Step 1: Examine the slope of the low-frequency asymptote of the plot.

↳ +20L dB/decade → Cascade of L identical differentiators, $(K_d s)^L$.

↳ –20L dB/decade → Cascade of L identical integrators, $(K_i/s)^L$.

↳ 0 dB/decade → DC gain, K_{dc} (without differentiator or integrator).

Step 2: Check for slope change while proceeding from low-frequency to high-frequency along the Bode magnitude plot.

↳ +20L dB/decade slope change at $\omega = \omega_o$ → Cascade of L identical zero factors, $(s/\omega_o + 1)^L$. Assuming all zeros are real.

↳ –20L dB/decade slope change at $\omega = \omega_o$ → L = 1 implies a pole factor $\frac{1}{s/\omega_o+1}$. L = 2 implies either a repeated pole factor $\left(\frac{1}{s/\omega_o+1}\right)^2$, or a 2nd order factor $\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$. L > 2 implies that there may be a hybrid of pole factors and 2nd order factors, $\left(\frac{1}{s/\omega_o+1}\right)^p \left(\frac{\omega_o^2}{s^2+2\zeta\omega_o s+\omega_o^2}\right)^q$, p + 2q = L.

Sampling & Reconstruction of Signals

Sampling is the conversion of an analog signal into a discrete–time sequence, which (after quantization) can be digitally stored and used.

Idealized Linear Time-Invariant Filters

Electronic filters are analog circuits used to remove undesired frequency components from a signal and/or enhancing desirable ones. The band of frequencies passed by a filter is called the pass-band while those rejected by a filter is called the stop-band. And idealized filter is one that has full transmission in the pass-band and complete attenuation in the stop-band.

Ideal Low–Pass Filter (LFP) – Noncausal

- Frequency Response: $H(f) = A\,\text{rect}(f/2B)$
- Impulse Response: $h(t) = 2AB\,\text{sinc}(2Bt)$
- Cutoff Frequency: Bandwidth = B (From 0 Hz to B Hz)

↳ Bandwidth of a system = Pass-band of its frequency response.

Ideal Band–Pass Filter (BPF) – Noncausal

- Frequency Response: $H(f) = A\left[\text{rect}\left(\frac{f+f_o}{B}\right) + \text{rect}\left(\frac{f-f_o}{B}\right)\right]$
- Impulse Response: $h(t) = 2AB\,\text{sinc}(Bt)\cos(2\pi f_o t)$
- Upper Cutoff Frequency: f_u
- Lower Cutoff Frequency: f_l
- Center Frequency: $f_o = 0.5(f_u + f_l)$
- Bandwidth: $B = f_u - f_l$

↳ Output magnitude gets multiplied with filter gain!

↳ Ideal filters are physically not realizable & can only be approximated by real–time filters. For example, the Butterworth filter.

Continuous–Time Sampling & Reconstruction of Lowpass Signals

- Sample by multiplying x(t) with a comb function of period T, which is equivalent to X(f) convoluting with a comb function of frequency 1/T & magnitude of 1/T in the frequency domain.
- Reconstruct by passing $x_s(t)$ through an ideal LPF, $x_s(t) * h(t)$, which is equivalent to $X_s(f) \times H(f)$.

Sampling Frequency $f_s = \frac{1}{T_s} < 2f_m$

↳ Spectral images overlap. This is called **frequency aliasing**.

↳ Perfect reconstruction is NOT possible.

Sampling Frequency $f_s = \frac{1}{T_s} > 2f_m$

↳ Gaps appear between spectral images due to **oversampling**.

↳ Perfect reconstruction is possible. Relaxes LPF design.

Conditions for Perfect Reconstruction

1. $x(t)$ MUST be bandlimited. $X(f) = 0; |f| > f_m$
2. Sampling rate of $x(t)$ $f_s \ge 2f_m$
3. Reconstruction LPF $|H(f)| = K\ if\ |f| > f_m$, $0\ if\ |f| > f_m$

↳ LPF should be ideal and have the same bandwidth as the signal.

Nyquist Sampling Frequency/Rate

Lowpass Signal: $2B$

Bandpass Signal: $2f_c + B$

Sampling Band-limited Bandpass Signal below Nyquist Rate

Two possible scenarios:

- Overlapping Spectral Images (Possible if $f_c > 0.5B$)

↳ $f_s = 2f_c/k$; $k = 1, 2, \cdots, \lfloor 2f_c/B \rfloor$

- Un-aliased Spectral Images (Possible if $f_c > 1.5B$)

↳ $\frac{2f_c+B}{k+1} \le f_s \le \frac{2f_c-B}{k}$ for $k = 1, 2, \ldots, \left\lfloor\frac{2f_c-B}{2B}\right\rfloor$

↳ Both formulas above are applicable for **symmetric** positive and negative spectral images. Overlapping spectral images will ALWAYS yield the lowest sampling frequency.

↳ When the positive and negative spectral images are **asymmetric**, only the un-aliased spectral image formula is applicable.

- For reconstruction, the ideal filter should be bandpass (usually rect functions at the positive and negative centre frequency with gain K).
- For overlapping spectral images, $K = 1/(2f_s)$ and for un-aliased spectral images, $K = 1/f_s$.
- When a bandpass signal is sampled at Nyquist Rate, an ideal lowpass filter (one big one covering both the negative and positive spectral images) may also be used to reconstruction but this will require higher sampling rate and also lead to introduction of noise.

Additional Information

1. Average Power (AP) = Total Energy (TE) in one period / period
2. Total energy for a signal can be calculated by breaking down the signal into piecewise functions, then squaring them and integrating them over their interval. Graphs such as rect or tri are easy to perform this technique.
3. To draw 1 – s or –s + 1, magnitude is same, phase is flipped (Negated)
4. Amplitude discrepancy for reconstructed signal is not distortion!
5. Time-Reversal in time domain will cause the frequency to flip sign only. Magnitude and phase stay the same! (Time-Scaling Property)
6. Amplitude scaling will cause both TE and AP to be multiplied by A^2.
7. Time Shift & Time Reversal have no effect on TE and AP.
8. Time Scaling: If signal is an energy signal (aperiodic), the TE scales by 1/|α|. AP = 0 for energy signals. If signal is a power signal (periodic), AP stays the same. TE = ∞ for power signals.
9. $sgn(-f) = -sgn(f)$
10. Average Power = Sum of $|c_k|^2$ (Magnitude of c_k! Not c_k!)
11. $X(0) = X(f)|_{f=0} = \int_{-\infty}^{\infty} x(t)\,dt$ (Area under $x(t)$, area can be negative if they are below x-axis!)
12. Any function in a convolution that is NOT bandlimited will cause result to be NOT bandlimited. $x(t)$ cannot be recovered from $x_s(t)$ without distortion.
13. When you add 2 signals and asked to find 1st–Null, take the LCM (intersection) of their frequency. ONLY works if the graph has no negative parts (Non-Negative Functions). LCM = Integer.
14. REAL signals have EVEN magnitude and ODD phase.
15. If c_k has a negative sign, NEED to change it to polar form and include in phase!
16. ALL ODD functions passes through origin.
17. If y = –4 from –∞ to 0, and y = 4 from 0 to ∞, differentiating this graph will give you an impulse of strength 8 at t = 0.
18. Average power of Cosine or Sine is $\frac{Amplitude\text{^}2}{2}$. Amplitude^2 for exponential & constants.
19. When 2 signals multiply, like a sinc^2 with a sinc, and you are asked to find the 1st–Null, you take their respective null frequency and **union** them. The lowest is your 1st–Null.
20. Degree of denominator polynomial in a transfer function determines the **order of the system**.
21. Time scaling do not affect C_k. Time shifting (t – a) introduces an exponential term $e^{j2\pi fa}$ to C_k. Time reversal causes the original C_k to be equivalent to the new C_{-k}. So, if C_6 = G, now C_{-6} = G. Amplitude scaling $A \cdot x(t)$ scales C_k by the new amplitude, A.
22. $x(t - t_0) = x(t) * \delta(t - t_0)$, use this formula if given $X(f)$ like sinc(f – 0.5) and ask to find $x(t)$.
23. For a **second-order system denominator**, it has roots $-\sigma \pm j\omega_d$. The undamped natural frequency: $\omega_n = \sqrt{\sigma^2 + \omega_d^2}$. $\sigma = \omega_n\zeta$.
24. **Smallest** damping ratio ⇒ poles as **close to imaginary axis** as allowed ⇒ real part |Re(s)| in s-plane as **small** as possible.
25. To have the **fastest/steepest response**, poles of the system should be as far left as possible in the s-plane.
26. No time operations will affect energy containment bandwidth or power containment bandwidth, but time-scaling ONLY. $x(at) \Rightarrow aB$.
27. Without initial zero conditions, $\mathcal{L}\{y(t)\} = Y(s)$, $\mathcal{L}\left\{\frac{dy(t)}{dt}\right\} = sY(s) - y(0^-)$, $\mathcal{L}\left\{\frac{d^2y(t)}{dt^2}\right\} = s^2Y(s) - s\,y(0^-) - y'(0^-)$.
28. Total average power = AC component + DC component (C_0).
29. Quadratic Formula $x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$
30. To Fourier Transform a periodic signal $\sum_{n=-\infty}^{\infty}\text{rect}\left(\frac{t-nT}{T_0}\right)$, first Fourier Transform a single pulse first (Take n = 0). The rect function will be transformed to a sinc function. **Periodic in time ⇒ impulses in frequency** at multiples of $f_s = 1/T$, with the same envelope. The final form will be a sum of impulses with the sinc function as its magnitude.
31. X multiplied with a Dirac-Comb that ranges from –∞ to ∞ is limited by the non-zero part of X. So, the number of impulses is finite instead.
32. When given a **time-shifted function** and asked to Laplace Transform, always Laplace Transform the original unshifted function first. Next, manipulate the terms (Addition/Subtraction or Multiplication/Division of extra terms) such that all t variables are shifted by the same amount.